Class 9: Candy Mini-Project

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Background

In this mini-project, we will explore FiveThirtyEight's Halloween Candy data set to find out what is the top ranked snack-sized Halloween candy? and What made some candies more desirable than others? Amongst other questions.

Importing Candy Data

First things first, let's get the data from the FiveThirtyEight GitHub repo. We will download this candy-data.csv file and place it in the project directory. Then, we will load it up with `read.csv()` and inspect the data to see exactly what we're dealing with.

```
candy <- read.csv("candy-data.txt", row.names = 1)

head(candy)
```

```
              chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand             1      0       1              0      0                1
3 Musketeers          1      0       0              0      1                0
One dime              0      0       0              0      0                0
One quarter           0      0       0              0      0                0
Air Heads             0      1       0              0      0                0
Almond Joy            1      0       0              1      0                0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

> Q1. How many different candy types are in this dataset?

There are 85 different candy types in this data set.

> Q2. How many fruity candy types are in the dataset?

```r
table(candy$fruity)
```

```

 0  1 
47 38 
```

There are 38 fruity candy types in the data set.

> Q3. What is your favorite candy (other than Twix) in the dataset and what is it's `winpercent` value?

```r
candy["Snickers", ]$winpercent
```

```
[1] 76.67378
```

> Q4. What is the `winpercent` value for "Kit Kat"?

```r
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the `winpercent` value for "Tootsie Roll Snack Bars"?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

Side Note: the skimr::skim() function There is a useful **`skim()`** function in the skimr package that can help give us a quick overview of a given data set. We will install this package and try it on our candy data.

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

> Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

The following columns look to be on a different scale to the majority of the other columns in the data set: sugarpercent, pricepercent, and winpercent.

> Q7. What do you think a zero and one represent for the candy$chocolate column?

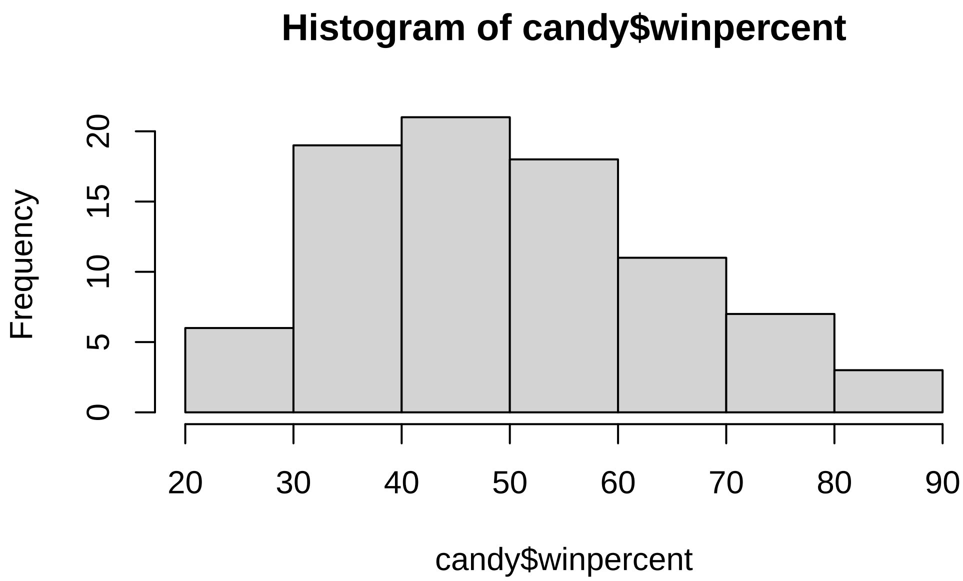
I think the zero and one represent how much of the data is missing and how much of the data is measured, respectively.

Exploratory Analysis

A good place to start any exploratory analysis is with a histogram.You can do this most easily with the base R function `hist()`. Alternatively, you can use `ggplot()` with `geom_hist()`

> Q8. Plot a histogram of `winpercent` values using both base R and ggplot2.

```r
hist(candy$winpercent)
```



Q9. Is the distribution of `winpercent` values symmetrical?

The distribution of winpercent values is not symmetrical.

Q10. Is the center of the distribution above or below 50%?

```
median(candy$winpercent)
```

```
[1] 47.82975
```

The center of distribution is below 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
choc.ind <- as.logical(candy$chocolate)
choc.candy <- candy[choc.ind, ]
choc.win <- choc.candy$winpercent
mean(choc.win)
```

```
[1] 60.92153
```

```r
fruity.ind <- as.logical(candy$fruity)
fruity.candy <- candy[fruity.ind, ]
fruity.win <- fruity.candy$winpercent
mean(fruity.win)
```

```
[1] 44.11974
```

On average chocolate candy is ranked higher than fruit candy.

> Q12. Is this difference statistically significant?

```r
t.test(choc.win, fruity.win)
```

```

	Welch Two Sample t-test

data:  choc.win and fruity.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y 
 60.92153  44.11974 
```

This difference is statistically significant.

Overall Candy Rankings

Let's use the base R `order()` function together with `head()` to sort the whole data set by a variable/column of interest.

> Q13. What are the five least liked candy types in this set?

```r
head(candy[order(candy$winpercent),], n=5)
```

```
                  chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                 0      1       0              0      0
Boston Baked Beans        0      0       0              1      0
Chiclets                  0      1       0              0      0
Super Bubble              0      1       0              0      0
Jawbusters                0      1       0              0      0
                  crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                        0    0   0        1        0.197        0.976
Boston Baked Beans               0    0   0        1        0.313        0.511
Chiclets                         0    0   0        1        0.046        0.325
Super Bubble                     0    0   0        0        0.162        0.116
Jawbusters                       0    1   0        1        0.093        0.511
                  winpercent
Nik L Nip           22.44534
Boston Baked Beans  23.41782
Chiclets            24.52499
Super Bubble        27.30386
Jawbusters          28.12744
```

The 5 least liked candies in this data set are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble and Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

```r
head(candy[order(-candy$winpercent),], n=5)
```

```
                         chocolate fruity caramel peanutyalmondy nougat
Reese's Peanut Butter cup        1      0       0              1      0
Reese's Miniatures               1      0       0              1      0
Twix                             1      0       1              0      0
Kit Kat                          1      0       0              0      0
Snickers                         1      0       1              1      1
                         crispedricewafer hard bar pluribus sugarpercent
Reese's Peanut Butter cup               0    0   0        0        0.720
Reese's Miniatures                      0    0   0        0        0.034
Twix                                    1    0   1        0        0.546
Kit Kat                                 1    0   1        0        0.313
Snickers                                0    0   1        0        0.546
                         pricepercent winpercent
Reese's Peanut Butter cup        0.651   84.18029
Reese's Miniatures               0.279   81.86626
Twix                             0.906   81.64291
```

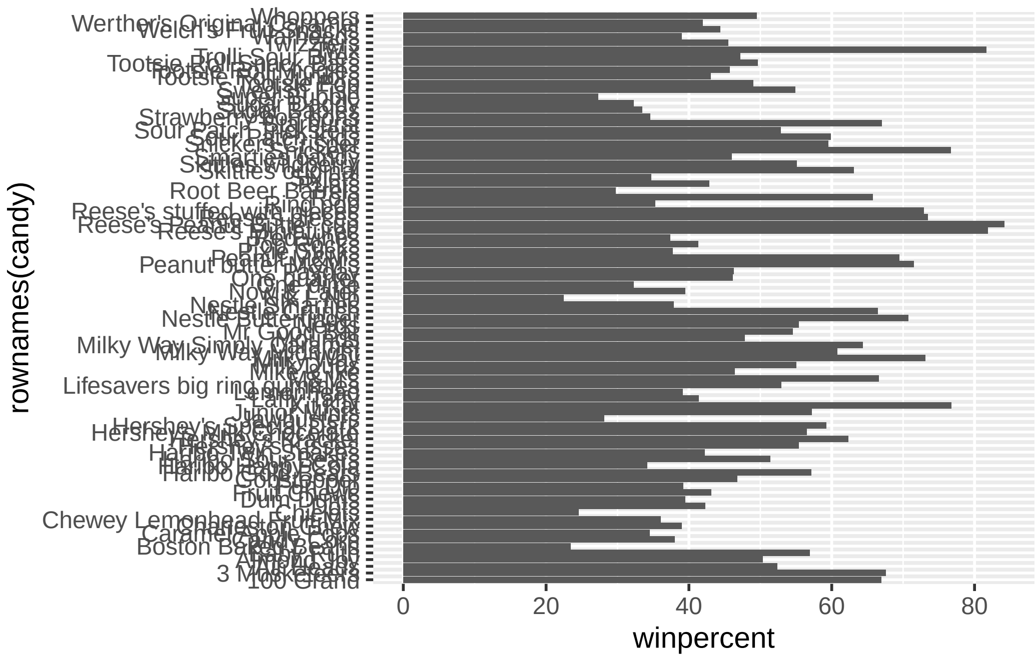
```
Kit Kat                           0.511   76.76860
Snickers                          0.651   76.67378
```

The 5 most liked candies in the data set are Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, and Snickers.

Q15. Make a first barplot of candy ranking based on winpercent values.

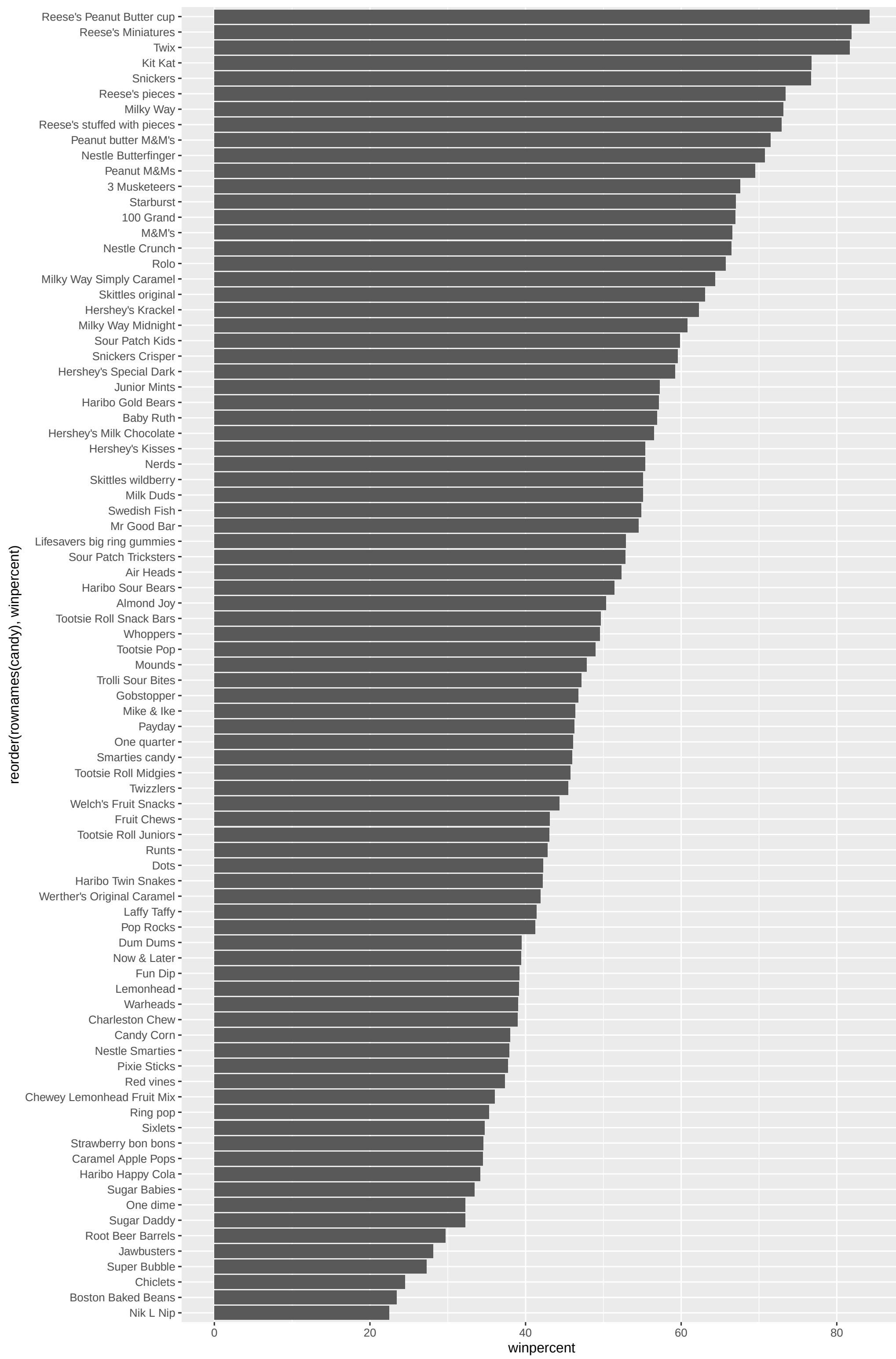
```r
library(ggplot2)

ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by `winpercent`?

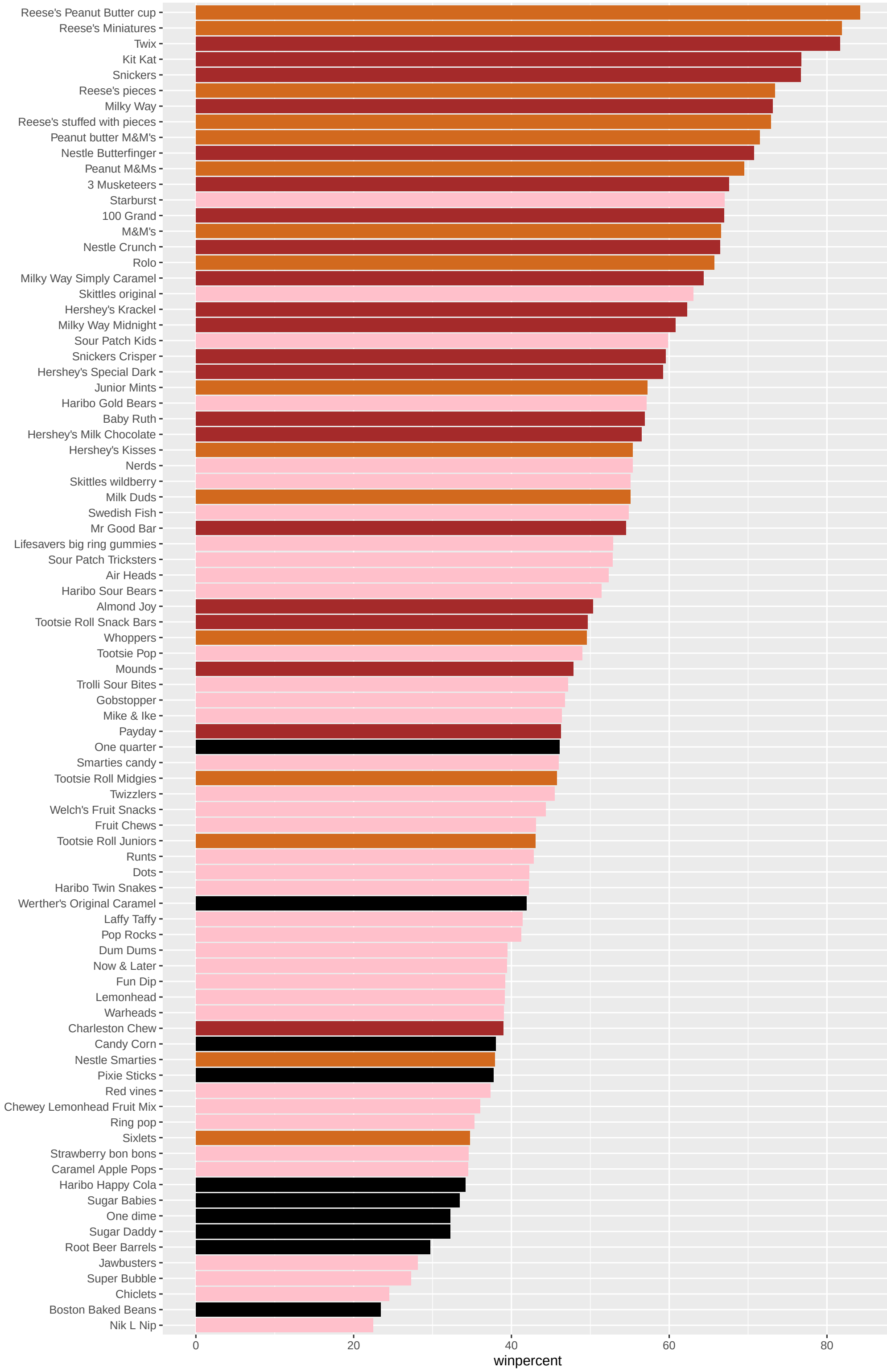
```r
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col()
```


Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip
reorder(rownames(candy), winpercent)
0
20
40
60
80
winpercent

Let's setup a color vector (that signifies candy type) that we can then use for some future plots.

```
mycols <- rep("black", nrow(candy))
mycols[as.logical(candy$chocolate)]="chocolate"
mycols[as.logical(candy$bar)]="brown"
mycols[as.logical(candy$fruity)]="pink"
```

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=mycols) +
  ylab("")
```


Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip
0
20
40
60
80
winpercent

Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy is Sixlets.

Q18. What is the best ranked fruity candy?

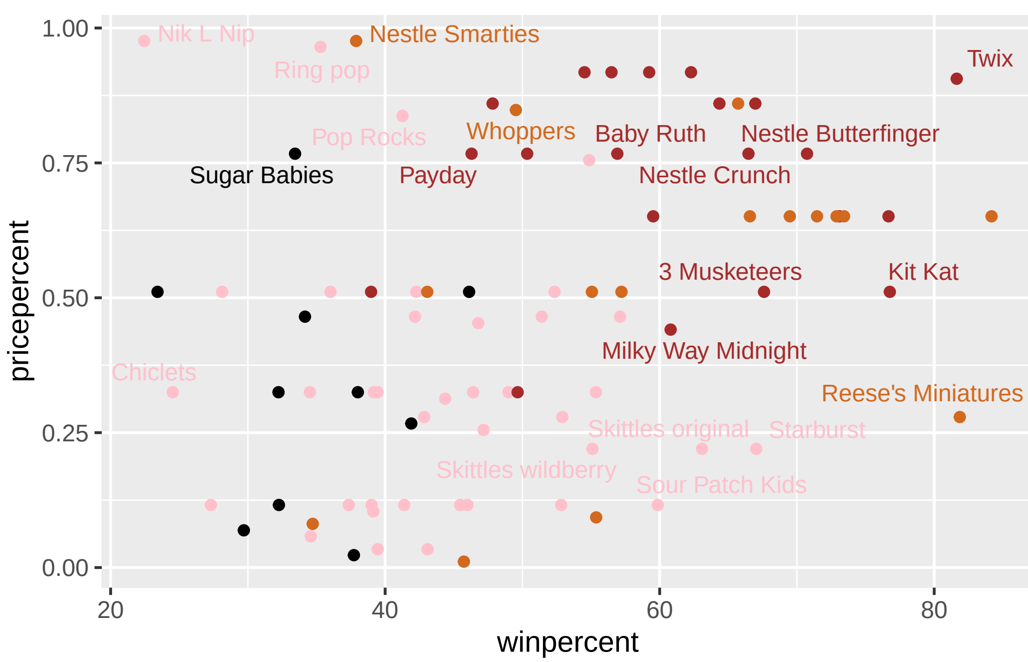
The best ranked fruity candy is Starburst.

Taking a look at pricepercent

What about value for money? What is the best candy for the least money? One way to get at this would be to make a plot of `winpercent` vs the `pricepercent` variable.

```r
library(ggrepel)

ggplot(candy) +
  aes(winpercent,
      pricepercent,
      label=rownames(candy)) +
  geom_point(col=mycols) +
  geom_text_repel(col=mycols, size = 3.3, max.overlaps=5)
```



Q19. What about value for money? What is the best candy for the least money?

```r
head(candy[order(candy$pricepercent),], n=3)
```

```
                     chocolate fruity caramel peanutyalmondy nougat
Tootsie Roll Midgies         1      0       0              0      0
Pixie Sticks                 0      0       0              0      0
Dum Dums                     0      1       0              0      0
                     crispedricewafer hard bar pluribus sugarpercent
Tootsie Roll Midgies                0    0   0        1        0.174
Pixie Sticks                        0    0   0        1        0.093
Dum Dums                            0    1   0        0        0.732
                     pricepercent winpercent
Tootsie Roll Midgies        0.011   45.73675
Pixie Sticks                0.023   37.72234
Dum Dums                    0.034   39.46056
```

The best candy for the least amount of money is Tootsie Roll Midgies.

> Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```r
head(candy[order(-candy$pricepercent),], n=5)
```

```
                         chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                        0      1       0              0      0
Nestle Smarties                  1      0       0              0      0
Ring pop                         0      1       0              0      0
Hershey's Krackel                1      0       0              0      0
Hershey's Milk Chocolate         1      0       0              0      0
                         crispedricewafer hard bar pluribus sugarpercent
Nik L Nip                               0    0   0        1        0.197
Nestle Smarties                         0    0   0        1        0.267
Ring pop                                0    1   0        0        0.732
Hershey's Krackel                       1    0   1        0        0.430
Hershey's Milk Chocolate                0    0   1        0        0.430
                         pricepercent winpercent
Nik L Nip                       0.976   22.44534
Nestle Smarties                 0.976   37.88719
Ring pop                        0.965   35.29076
Hershey's Krackel               0.918   62.28448
Hershey's Milk Chocolate        0.918   56.49050
```

The top 5 most expensive candy types are Nestle Smarties, Ring Pop, Nik L Nip, Hershey's Krackel, and Hershey's Milk Chocolate. The least popular amongst all of these is Nik L Nip.

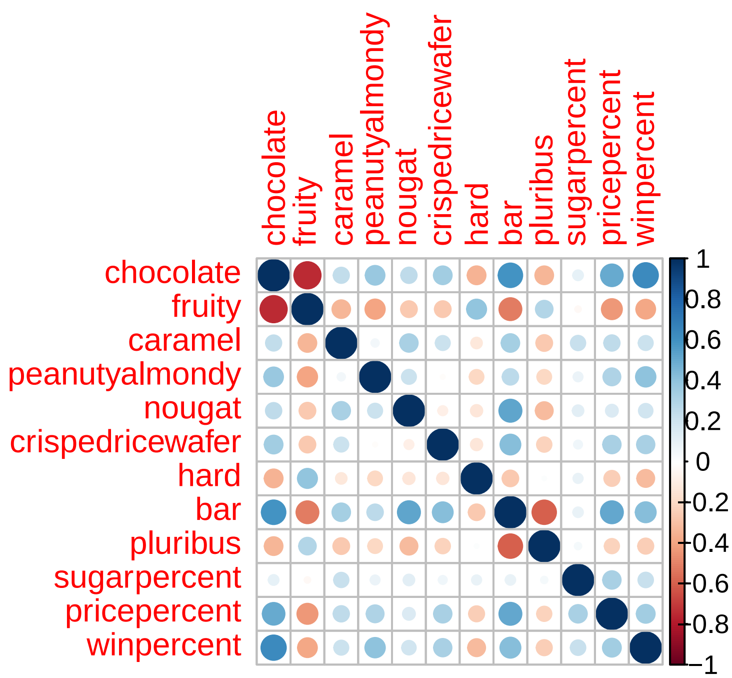
Exploring the Correlation Structure

Before we dive into PCA in the next section, it's worth pausing to look more closely at the correlations between all pairs of variables.

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cij <- cor(candy)
corrplot(cij)
```



> Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

The two variables that are anti-correlated are fruity and chocolate, fruity and bar, pluribus and bar.

> Q23. Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

The variables we could expect to have the largest contributions to PC1 are fruity and chocolate.

Principal Component Analysis

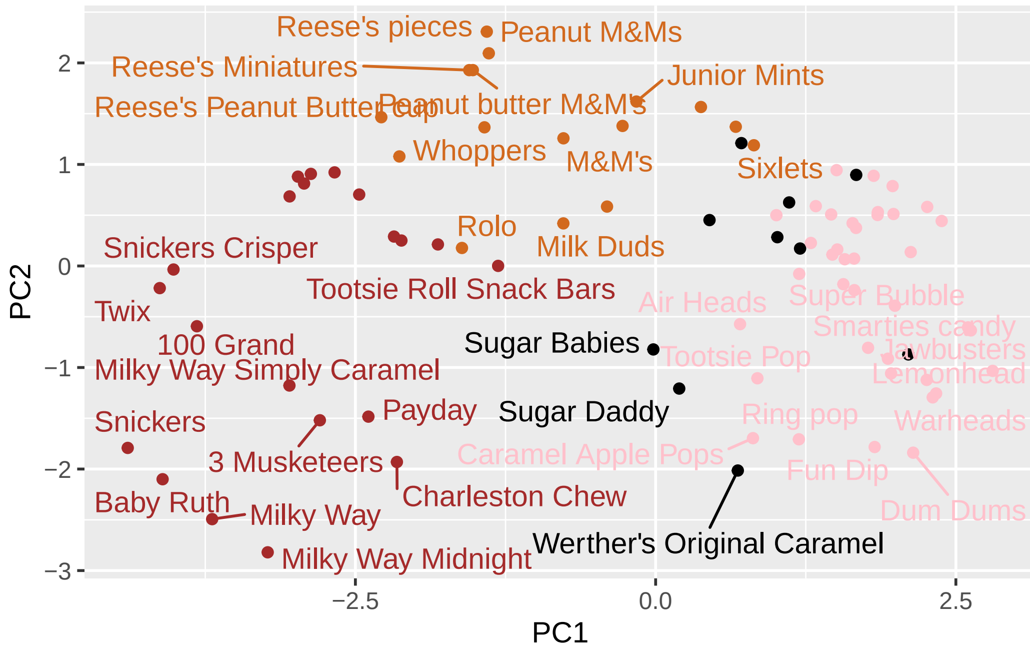
In this case we want to be sure to set `scale = TRUE` argument for `prcomp()` because we have one variable `winpercent` that is on a very different scale than all others and would otherwise dominate our PCA results

```
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

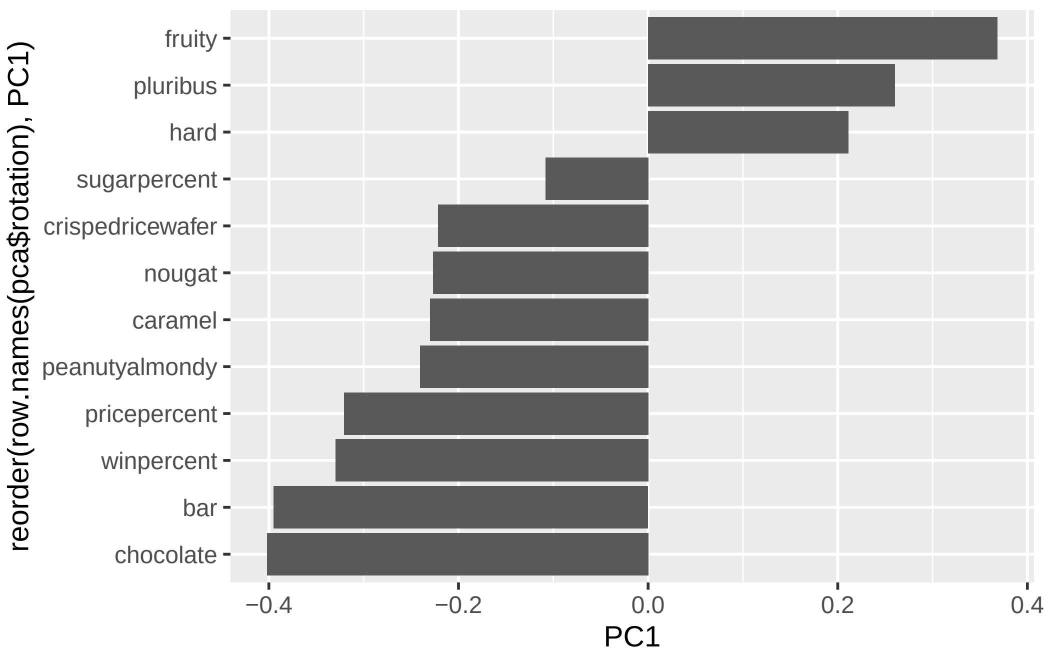
```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

First major result figure is the "score plot" of PC1 vs PC2 - how the different candy are related to each other on our new PC axis:

```
ggplot(pca$x) +
  aes(PC1, PC2, label=row.names(pca$x))+
  geom_point(col=mycols)+
  geom_text_repel(col=mycols)
```




```r
ggplot(pca$rotation) +
  aes(PC1,
      reorder(row.names(pca$rotation), PC1)) +
  geom_col()
```



> Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The original variables that are picked up strongly by PC1 in the positive direction are fruity, pluribus, and hard. This makes sense as we saw in the corrplot that there is no correlation between fruity and chocolate therefore the divide in the data would be significant.

Summary

> Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a "winning" candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

The following combination of characteristics appears to make a "winning" candy: chocolate, bar, peanuts, and nougat.

The plots generated with the data help identify these findings as they show us multiple times that there is no correlation between fruity and chocolate (corrplot) and also visually demonstrate that the most popular candy type between chocolate and fruity is in fact chocolate (Visualization).